

Analysis of arthropod communities in Hawaii using METE Theory

Setup

```
library(meteR)  
library(ggplot2)  
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(tidyr)  
library(vegan)
```

Loading required package: permute

```
library(HawaiiArthMETE)
```

Attaching package: 'HawaiiArthMETE'

The following object is masked from 'package:meteR':

arth

```
# load data
data(arth)
```

METE calculations

```
arth_mete <- split(arth, arth[, c("site", "tree")], drop = TRUE) |>
  lapply(function(x) {
    cat("\n", "site: ", x$site[1], ", tree:", x$tree[1], "\n", sep = "")
    # the esf for this grouping
    esf <- meteESF(spp = x$species_code,
                  abund = x$abundance,
                  power = x$metabolic_rate)

    # sad and ipd
    s <- sad(esf)
    p <- ipd(esf)

    data.frame(
      site = x$site[1],
      tree = x$tree[1],

      # z2 values
      z2_sad = logLikZ(s)$z,
      z2_ipd = logLikZ(p)$z,

      # prop non-native for abundance and power
      prop_n_non_nat = sum(
        x$abundance[!(x$origin %in% c("endemic", "indigenous"))]
      ) /
        sum(x$abundance),
      prop_p_non_nat = sum(
        x$metabolic_rate[!(x$origin %in% c("endemic", "indigenous"))]
      ) /
        sum(x$metabolic_rate),

      # differences in extreme values
```

```

        n1_ratio = mean(meteDist2Rank(s) == 1) - mean(s$data == 1),
        nmax_ratio = sum(meteDist2Rank(s)[1:3]) - sum(s$data[1:3]),
        p1_ratio = mean(meteDist2Rank(p) <= 2) - mean(p$data <= 2),
        pmax_ratio = sum(meteDist2Rank(p)[1:3]) - sum(p$data[1:3])
    )
})

```

```

site: 2, tree:0
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

```

```

site: 1, tree:1
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

```

```

site: 2, tree:1
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

```

```

site: 3, tree:1
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

```

```

site: 4, tree:1
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

```

```

site: 5, tree:1
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:

```

attempt 1

site: 1, tree:2
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 2, tree:2
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 3, tree:2
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 4, tree:2
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 5, tree:2
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 1, tree:3
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 3, tree:3
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 4, tree:3
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 5, tree:3
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 1, tree:4
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 2, tree:4
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 3, tree:4
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 4, tree:4
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 5, tree:4
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 1, tree:5
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 2, tree:5
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 3, tree:5
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 5, tree:5
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 1, tree:6
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 2, tree:6
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 3, tree:6
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 5, tree:6

simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 1, tree:7
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 2, tree:7
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 3, tree:7
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 5, tree:7
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 1, tree:8
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 2, tree:8
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 3, tree:8
simulating data that conform to state variables:

attempt 1
simulating data that conform to state variables:
attempt 1

site: 5, tree:8
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 1, tree:9
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 2, tree:9
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 3, tree:9
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 5, tree:9
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 1, tree:10
simulating data that conform to state variables:
attempt 1
simulating data that conform to state variables:
attempt 1

site: 2, tree:10
simulating data that conform to state variables:
attempt 1


```
simulating data that conform to state variables:  
attempt 1
```

```
site: 3, tree:10  
simulating data that conform to state variables:  
attempt 1  
simulating data that conform to state variables:  
attempt 1
```

```
site: 3, tree:11  
simulating data that conform to state variables:  
attempt 1  
simulating data that conform to state variables:  
attempt 1
```

```
site: 4, tree:11  
simulating data that conform to state variables:  
attempt 1  
simulating data that conform to state variables:  
attempt 1
```

```
site: 4, tree:12  
simulating data that conform to state variables:  
attempt 1  
simulating data that conform to state variables:  
attempt 1
```

```
site: 4, tree:13  
simulating data that conform to state variables:  
attempt 1  
simulating data that conform to state variables:  
attempt 1
```

```
site: 4, tree:15  
simulating data that conform to state variables:  
attempt 1  
simulating data that conform to state variables:  
attempt 1
```

```
arth_mete <- do.call(rbind, arth_mete)
```

What are the highly invaded, highly dominated samples?

```
outliers <- filter(arth_mete, nmax_ratio < -300)
```

Two samples stand out as clear outliers in terms of deviation from SAD predictions as well as being the two most highly invaded samples. The two samples come from the Laupāhoehoe and Kohala sites. To further understand these two samples and determine if they should remain in the analyses we look at their composition. Table 1 shows the three most abundant species for the Laupāhoehoe outlier sample while Table 2 shows the three most abundant species for the Kohala outlier sample.

```
filter(arth, tree == outliers$tree[1] & site == outliers$site[1]) |>
  summarize(n = n(), .by = c(genus, species, origin)) |>
  arrange(desc(n)) |>
  head(3) |>
  knitr::kable(
    caption = paste(
      "Top three most abundant taxa from",
      levels(arth$site_name)[as.numeric(outliers$site[1])],
      "outlier sample"
    )
  )
```

Table 1: Top three most abundant taxa from Laupāhoehoe outlier sample

genus	species	origin	n
Greenidea	formosana	adventive	423
Salina	celebensis	adventive	330
Entomobrya	socia	adventive	69

```
filter(arth, tree == outliers$tree[2] & site == outliers$site[2]) |>
  summarize(n = n(), .by = c(genus, species, origin)) |>
  arrange(desc(n)) |>
  head(3) |>
  knitr::kable(
    caption = paste(
      "Top three most abundant taxa from",
      levels(arth$site_name)[as.numeric(outliers$site[2])],
      "outlier sample"
    )
  )
```

Table 2: Top three most abundant taxa from Kohala outlier sample

genus	species	origin	n
Salina	celebensis	adventive	410
Forcipomyia	hardyi	endemic	92
Kilauella	sp_3	endemic	46

```
arth_mete_out <- arth_mete
arth_mete <- filter(arth_mete,
                    !((site %in% outliers$site) & (tree %in% outliers$tree)))
```

```
filter(arth, tree == outliers$tree[1] & site == outliers$site[1]) |>
  summarize(n = n(), .by = c(genus, species, origin)) |>
  arrange(desc(n)) |>
  head(3)
```

```
filter(arth, tree == outliers$tree[1] & site == outliers$site[1]) |>
  summarize(n = n(), .by = c(tray, genus, species, origin)) |>
  arrange(tray) |>
  tidyr::pivot_wider(names_from = tray, values_from = n) |>
  View()
```

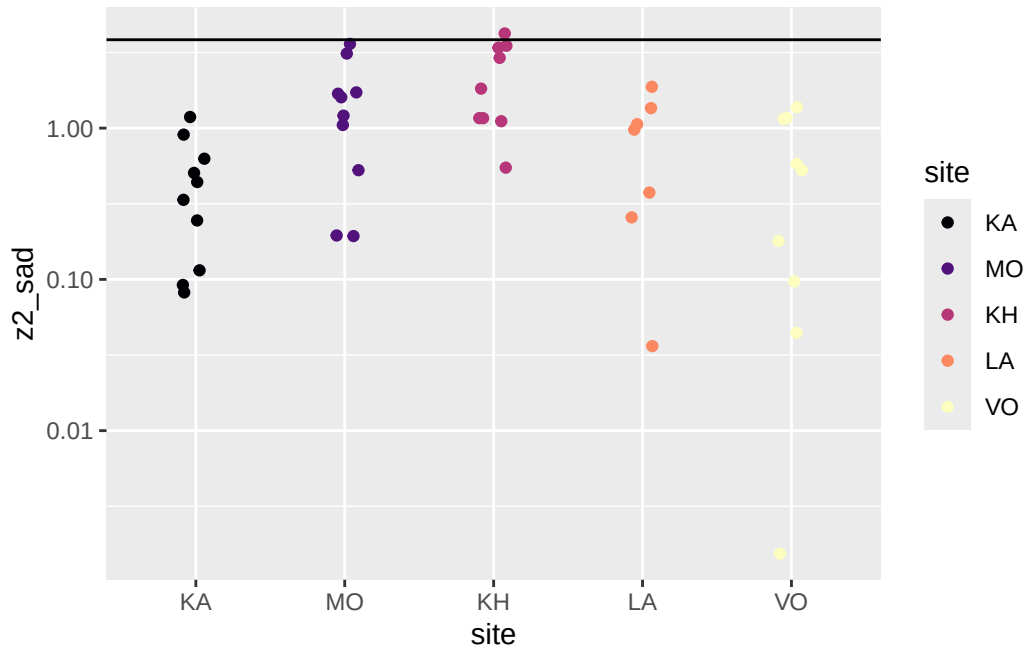
```
filter(arth, tree == outliers$tree[2] & site == outliers$site[2]) |>
  summarize(n = n(), .by = c(tray, genus, species, origin)) |>
  arrange(tray) |>
  tidyr::pivot_wider(names_from = tray, values_from = n) |>
  View()
```

```
filter(arth, tree == outliers$tree[2] & site == outliers$site[2]) |>
  summarize(n = n(), .by = c(genus, species, origin)) |>
  arrange(desc(n)) |>
  head()
```

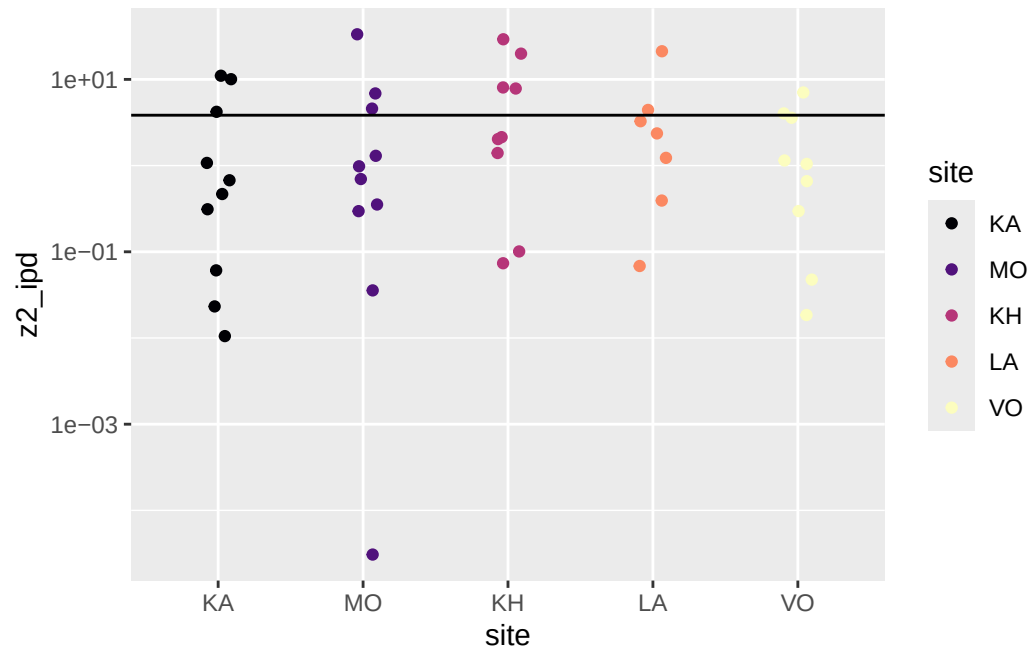
```
filter(arth, genus == "Salina" & species == "celebensis") |>
  summarize(n = n(), .by = c(site, tree, tray)) |>
  arrange(desc(n)) |>
  head()
```

Visualizing deviations from METE across chronosequence

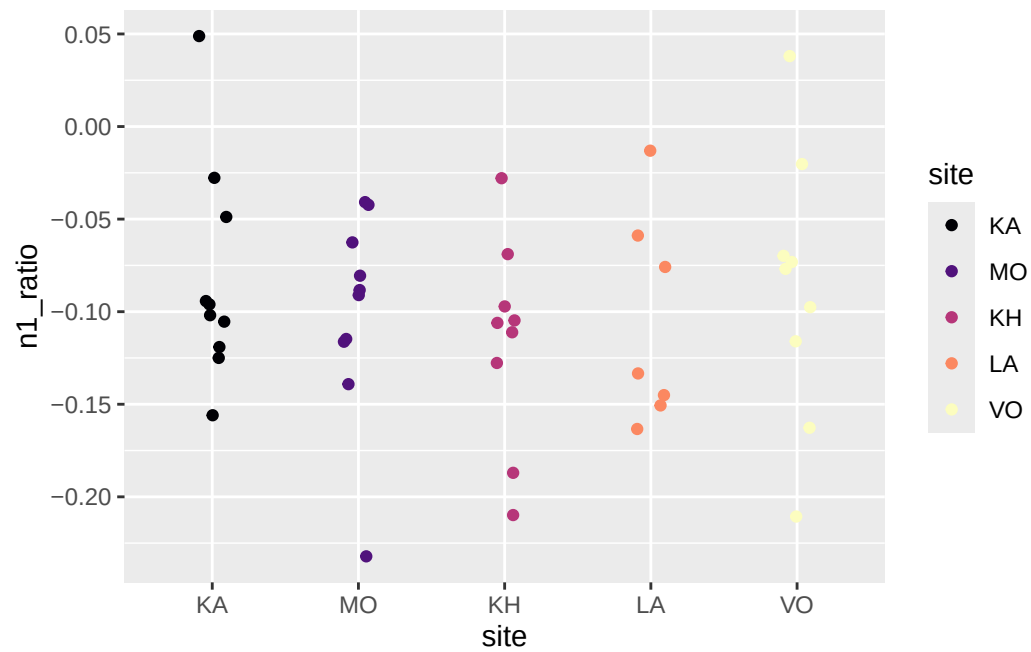
```
ggplot(arth_mete, aes(site, z2_sad, color = site)) +  
  geom_jitter(width = 0.1) +  
  scale_y_log10() +  
  scale_color_viridis_d(option = "magma") +  
  geom_hline(yintercept = qchisq(0.95, 1))
```



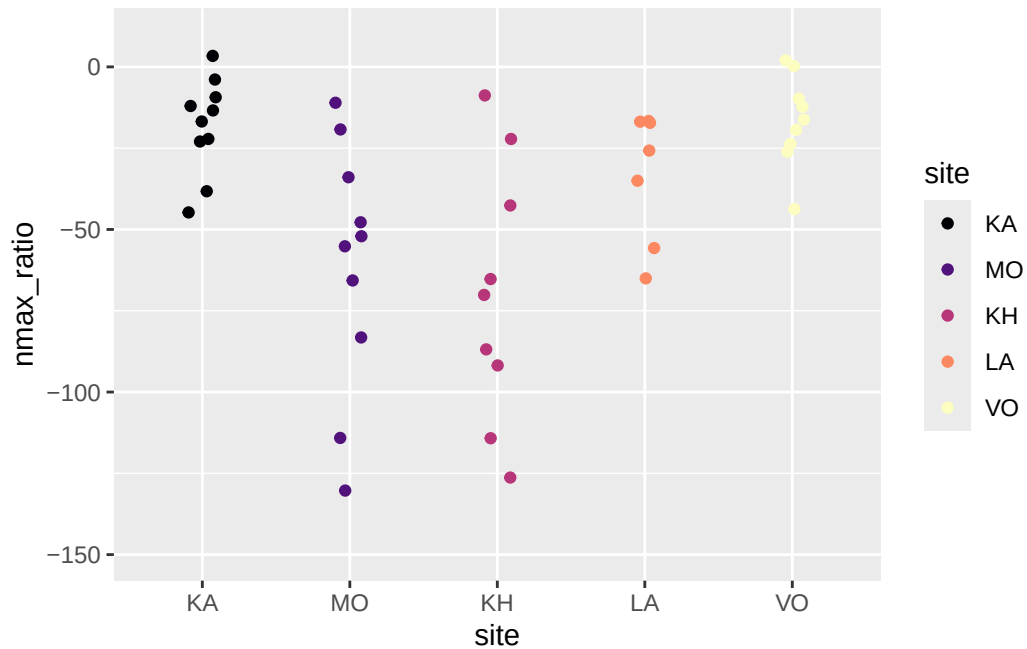
```
ggplot(arth_mete, aes(site, z2_ipd, color = site)) +  
  geom_jitter(width = 0.1) +  
  scale_y_log10() +  
  scale_color_viridis_d(option = "magma") +  
  geom_hline(yintercept = qchisq(0.95, 1))
```



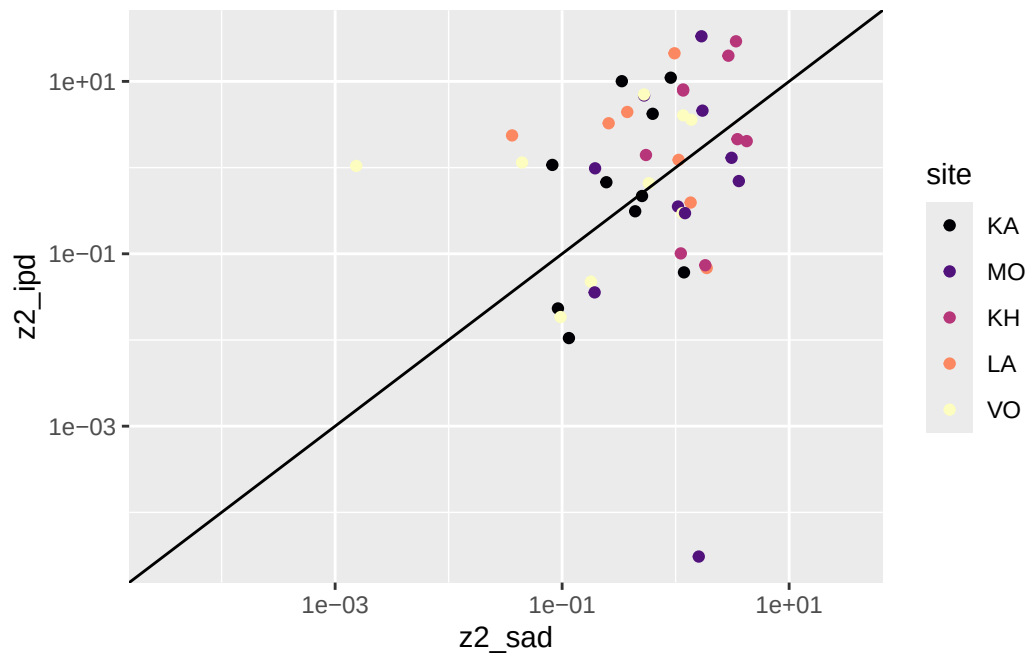
```
ggplot(arth_mete, aes(site, n1_ratio, color = site)) +
  geom_jitter(width = 0.1) +
  scale_color_viridis_d(option = "magma")
```



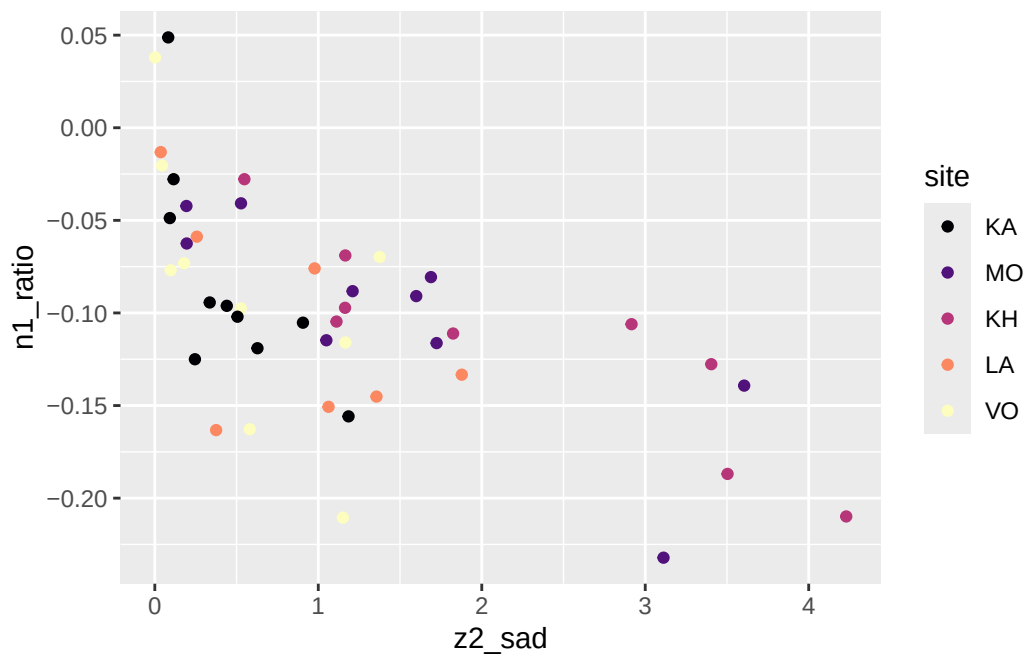
```
ggplot(arth_mete, aes(site, nmax_ratio, color = site)) +
  geom_jitter(width = 0.1) +
  scale_color_viridis_d(option = "magma") +
  ylim(c(-150, 10))
```



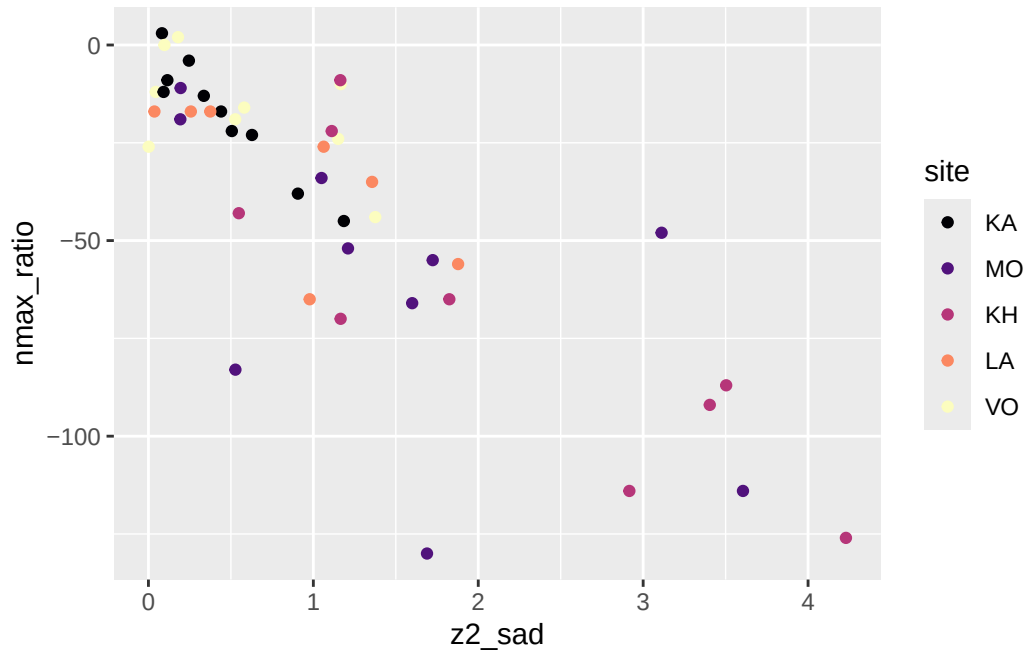
```
ggplot(arth_mete, aes(z2_sad, z2_ipd, color = site)) +
  geom_point() +
  scale_x_log10(limits = range(select(arth_mete, starts_with("z2")))) +
  scale_y_log10(limits = range(select(arth_mete, starts_with("z2")))) +
  scale_color_viridis_d(option = "magma") +
  geom_abline(slope = 1, intercept = 0)
```

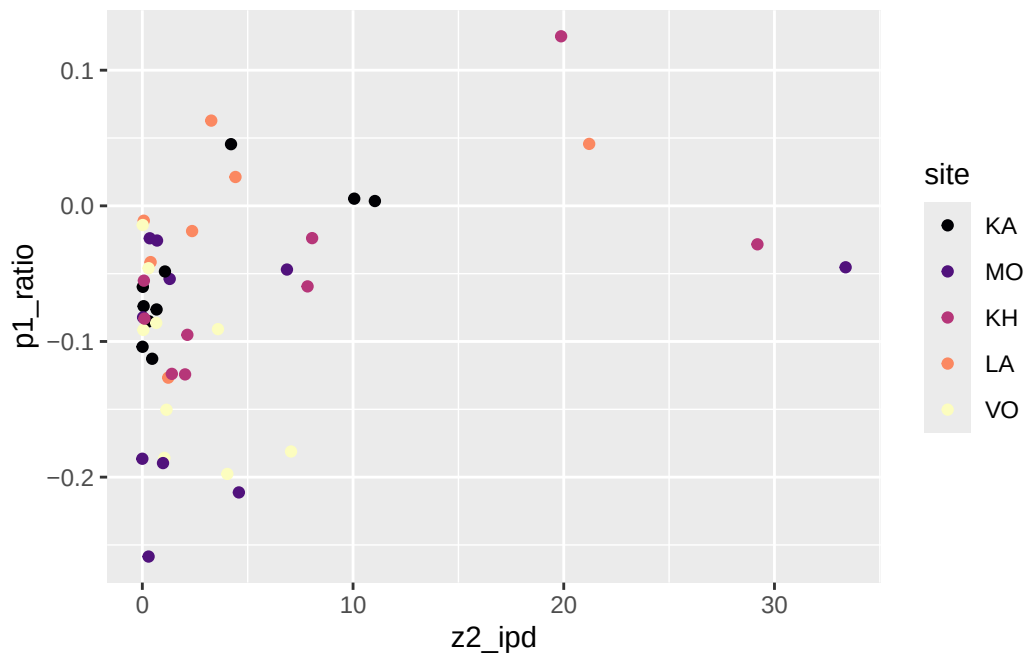


```
ggplot(arth_mete, aes(z2_sad, n1_ratio, color = site)) +
  geom_point() +
  scale_color_viridis_d(option = "magma")
```

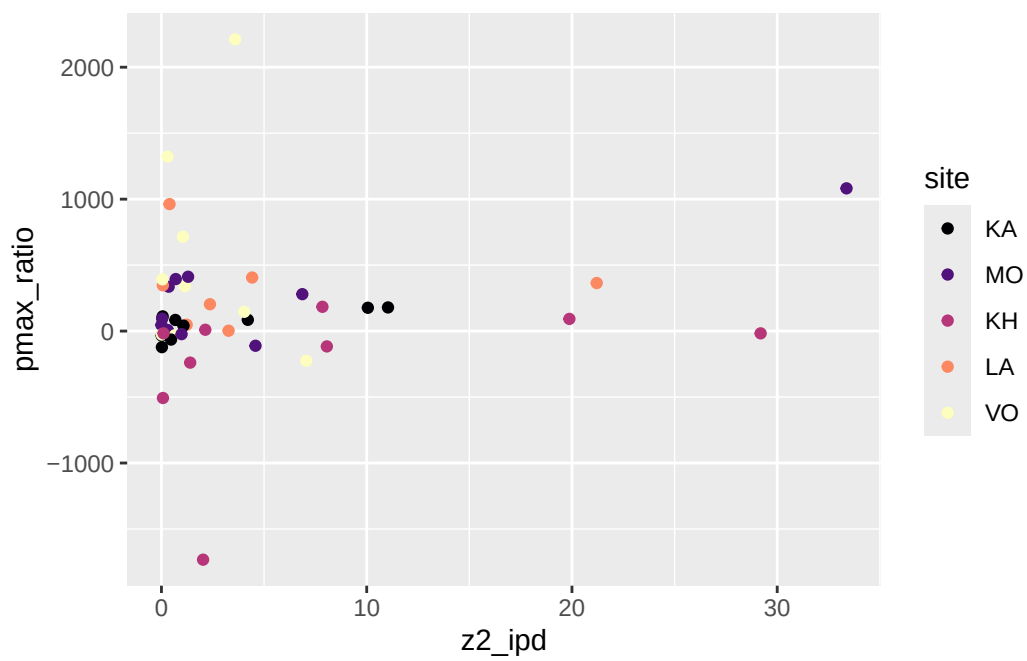


```
ggplot(arth_mete, aes(z2_sad, nmax_ratio, color = site)) +
  geom_point() +
  scale_color_viridis_d(option = "magma")
```





```
ggplot(arth_mete, aes(z2_ipd, pmax_ratio, color = site)) +
  geom_point() +
  scale_color_viridis_d(option = "magma")
```



How do METE and the real data differ

Which one over-predicts rarity?

Comparing deviations from METE with potential explanatory variables

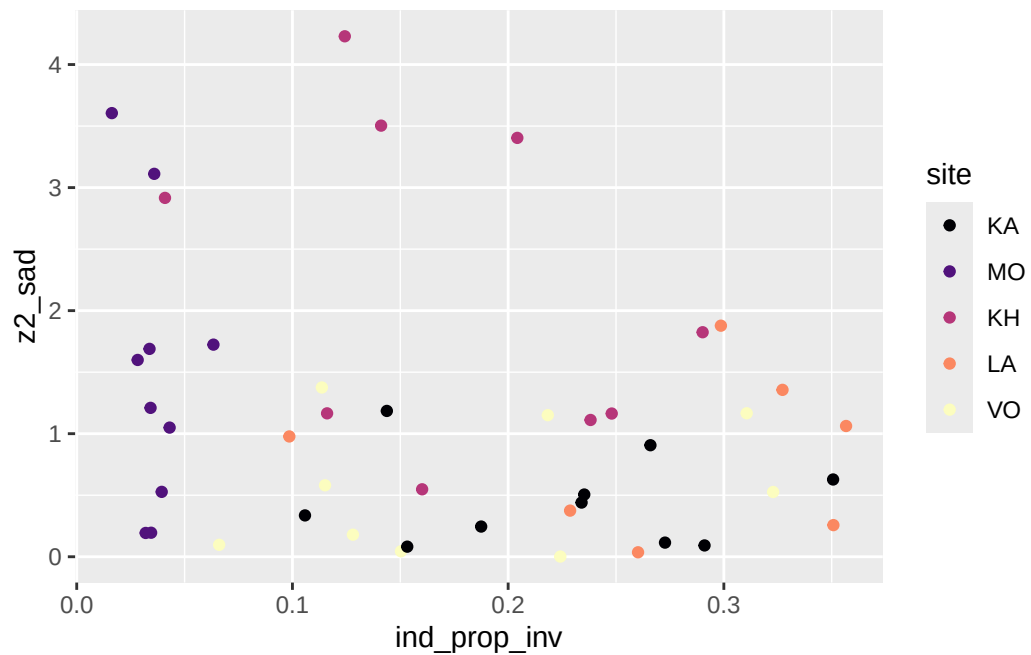
Deviation vs. proportion that are invasive

```
arth_inv <- group_by(arth, site, tree) |>
  summarize(
    spp_prop_inv = n_distinct(
      species_code[origin %in% c("adventive", "intentional introduction")]
    ) /
    n_distinct(species_code),
    ind_prop_inv = mean(origin %in% c("adventive", "intentional introduction")),
    met_prop_inv = sum(
      metabolic_rate[origin %in% c("adventive", "intentional introduction")]
    ) /
    sum(metabolic_rate)
  ) |>
  ungroup()
```

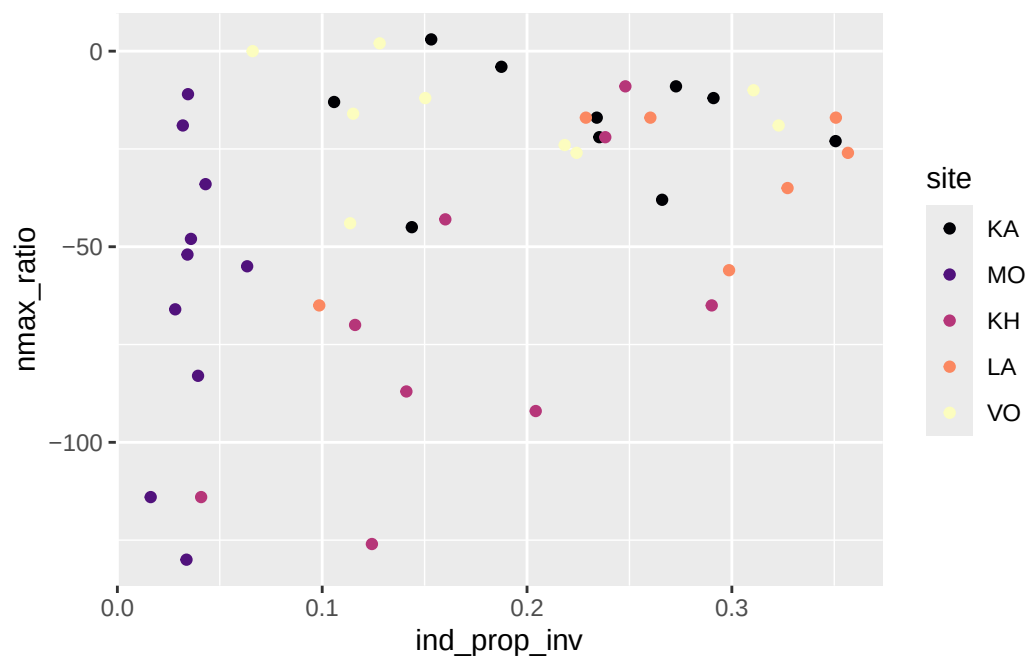
`summarise()` has regrouped the output.
i Summaries were computed grouped by site and tree.
i Output is grouped by site.
i Use `summarise(.groups = "drop\_last")` to silence this message.
i Use `summarise(.by = c(site, tree))` for per-operation grouping
(`?dplyr::dplyr\_by`) instead.

```
mete_by_inv <- left_join(arth_mete, arth_inv, by = c("site", "tree"))

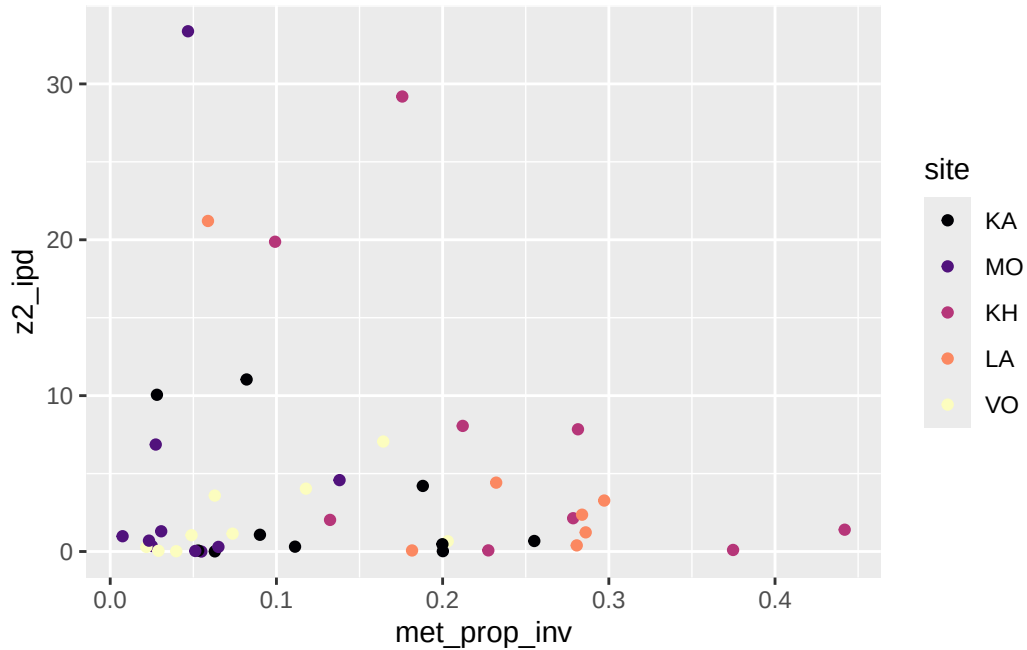
ggplot(mete_by_inv, aes(ind_prop_inv, z2_sad, color = site)) +
  geom_point() +
  scale_color_viridis_d(option = "magma")
```



```
ggplot(mete_by_inv, aes(ind_prop_inv, nmax_ratio, color = site)) +
  geom_point() +
  scale_color_viridis_d(option = "magma")
```



```
ggplot(mete_by_inv, aes(met_prop_inv, z2_ipd, color = site)) +
  geom_point() +
  scale_color_viridis_d(option = "magma")
```



Deviation vs. dissimilarity

```
z_diss <- function(x, B = 500) {
  obs_dist <- x |> vegdist() |> mean()

  sim <- nullmodel(x, "quasiswap_count") |>
    simulate(nsim = B)

  sim_mean_dist <- sapply(1:dim(sim)[3], function(i) {
    vegdist(sim[, , i]) |> mean()
  })

  (obs_dist - mean(sim_mean_dist)) / sd(sim_mean_dist)
}
```

```

diss <- split(arth, arth$site) |>
  sapply(FUN = function(dat) {
    summarize(dat, abund = n(), .by = c(tree, species_code)) |>
      pivot_wider(names_from = species_code,
                  values_from = abund,
                  values_fill = 0) |>
      select(!tree) |>
      z_diss()
  })

mete_by_diss <- mutate(arth_mete,
                      mean_diss = diss[site])

summarize(mete_by_diss, .by = site,
          mean_diss = first(mean_diss)) |>
ggplot(aes(site, mean_diss, color = site)) +
  geom_point() +
  scale_color_viridis_d(option = "magma")

ggplot(mete_by_diss, aes(mean_diss, z2_sad, color = site)) +
  geom_jitter(width = 0.25) +
  scale_color_viridis_d(option = "magma")

```